



Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

## GROUP 'A' [MCQ] [11×3=33]

- The mixture of solution having pH lesser than 7 is  
a)  $\text{NH}_4\text{OH} + \text{NH}_4\text{Cl}$  b)  $\text{Cu}(\text{OH})_2 + \text{CuSO}_4$   
c)  $\text{CH}_3\text{COOH} + \text{CH}_3\text{COONa}$  d)  $\text{Fe}(\text{OH})_3 + \text{FeCl}_3$
- The normality of orthophosphoric acid having purity of 70% by weight and specific gravity 1.54 would be  
a) 11 N b) 22 N c) 33 N d) 44 N
- Example of primary standard substance is  
a) oxalic acid crystal b)  $\text{KMnO}_4$   
c)  $\text{HCl}$  d)  $\text{H}_2\text{SO}_4$
- Common ion effect is not applicable for  
a)  $\text{HCl} + \text{H}_2\text{SO}_4$  b)  $\text{H}_2\text{S} + \text{HCl}$   
c)  $\text{NH}_4\text{OH} + \text{NH}_4\text{Cl}$  d)  $\text{CH}_3\text{COOH} + \text{CH}_3\text{COONa}$
- The compound whose 0.1M solution is neutral is  
a)  $\text{NH}_4\text{Cl}$  b)  $\text{CH}_3\text{COONa}$   
c)  $(\text{NH}_4)_2\text{CO}_3$  d)  $(\text{NH}_4)_2\text{SO}_4$
- Best reagent to convert alcohol into alkyl chloride is  
a)  $\text{PCl}_3$  b)  $\text{SOCl}_2$   
c)  $\text{PCl}_5$  d)  $\text{HCl/ZnCl}_2$
- The most common reaction that alkyl halide shows is  
a) Electrophilic addition b) Nucleophilic substitution  
c) Free radical addition d) Electrophilic substitution
- Which is liquid at room temperature?

- a)  $\text{CH}_3\text{F}$  b)  $\text{CH}_3\text{Cl}$  c)  $\text{CH}_3\text{I}$  d)  $\text{CH}_3\text{Br}$

9. Which metal is alloyed with copper to form bronze?

- a) Fe b) Sn  
c) Zn d) Pb

10. Schweitzer's reagent is

- a)  $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$  b)  $\text{Cu}_2\text{SO}_4 \cdot \text{H}_2\text{O}$   
c)  $\text{Cu}(\text{NH}_4)_2\text{SO}_4 \cdot 4\text{H}_2\text{O}$  d)  $\text{Cu}(\text{NH}_3)_4\text{SO}_4$

11. Spelter is

- a)  $\text{ZnS}$  b)  $\text{ZnO}$  c) impure zinc d) pure zinc

## GROUP 'B' [E-S=40]

Short Questions:

- Calculate the equivalent weight of  $\text{KMnO}_4$  in various medium. [3]
- 1.5 g of  $\text{CaCO}_3$  sample required 40ml of  $\text{N}/2$   $\text{HCl}$  for complete neutralization. Calculate the percentage purity of  $\text{CaCO}_3$  in the sample. [2]
- Define Redox titration with suitable example. [2]
- Calculate the pH of mixture solution obtained by mixing 200ml of solution having  $\text{pH}=2$  and 300 ml of another solution having  $\text{pH}=12$ . [3]
- Water is Lewis base and Bronsted acid, Explain. [2]
- Calculate the pH of  $10^{-4}$  M  $\text{NaOH}$  solution. [2]
- Why is aqueous solution of  $\text{Na}_2\text{CO}_3$  basic in nature? [1]
- Write short note on "Chemistry of Blue vitriol". [3]
- What is blister copper? Explain. [2]
- Explain why compounds of zinc are colourless? [1]
- Mention the possible oxidation number of Mn. [1]
- Most of the transition elements show paramagnetic behavior, why? [1]
- $\text{Fe}^{3+}$  is more stable than  $\text{Fe}^{2+}$ , why? [1]
- Explain why  $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$  ion is coloured? [1]



GROUP 'A' [MCQ] [11×1=11]

1. Common salt contains 50 ppm iodine in 1kg packet. Amount of iodine in 1 kg packet salt is

- a)  $5 \times 10^{-5} \text{ Kg}$  b)  $5 \times 10^{-4} \text{ Kg}$   
c)  $5 \times 10^{-6} \text{ Kg}$  d)  $5 \times 10^{-3} \text{ Kg}$

2. The highest pH value would be shown by a solution of 0.1 M solution of

- a)  $\text{H}_2\text{SO}_4$  b) NaOH c)  $\text{Ca}(\text{OH})_2$  d)  $\text{Na}_2\text{CO}_3$

3. Conjugate base of strong acid is

- a) stable b) unstable  
c) reactive d) metastable

4. Which of the following can act as a base according to the Lewis concept?

- a) H b) K c)  $\text{Na}^+$  d) Cl

5. Maximum Oxidation state of Ti is

- a) +2 b) +3  
c) +4 d) +7

6. Ruby copper is

- a)  $\text{Cu}_2\text{O}$  b)  $\text{Cu}_2\text{S}$   
c)  $\text{Cu}(\text{OH})_2$  d)  $\text{Cu}(\text{OH})_2 \cdot \text{CuCO}_3$

7. Blast furnace cannot be used for the extraction of zinc because

- a) low melting point of zinc b) low boiling point of zinc  
c) high reactivity of zinc d) high malleability of zinc

8. Relative stability of carbocation is in the order.

- a)  $3^\circ > 2^\circ > 1^\circ$  b)  $2^\circ > 3^\circ > 1^\circ$   
c)  $1^\circ > 2^\circ > 3^\circ$  d)  $2^\circ > 1^\circ > 3^\circ$

9. Hydrolysis of alkyl halide with aqueous caustic alkali is

- a) electrophilic solution b) nucleophilic substitution  
c) Free radical substitution d) nucleophilic addition

10. Number of isomeric dichlorobenzene are

- a) 2 b) 3 c) 5 d) 7

11.  $\text{SN}_1$  reaction involves formation of

- a) carbocation b) carbanion c) free radical d) carbene

GROUP 'B' [8×5=40]

Short Questions

1. a) Calculate the equivalent weight of  $\text{KMnO}_4$  in various medium. [3]

b) 1.5 g of  $\text{CaCO}_3$  sample required 40ml of  $\text{N}/2$  HCl for complete neutralization. Calculate the percentage purity of  $\text{CaCO}_3$  in the sample. [2]

2. a) Define Redox titration with suitable example. [2]

b) Calculate the pH of mixture solution obtained by mixing 200ml of solution having  $\text{pH}=2$  and 300 ml of another solution having  $\text{pH}=12$ . [3]

3. a) Water is Lewis base and Bronsted acid. Explain. [2]

b) Calculate the pH of  $10^{-2} \text{ M}$  NaOH solution. [2]

c) Why is aqueous solution of  $\text{Na}_2\text{CO}_3$  basic in nature? [1]

4. a) Write short note on 'Chemistry of Blue vitriol'. [3]

b) What is blister copper? Explain. [2]

5. a) Explain why compounds of zinc are colourless? [1]

b) Mention the possible oxidation number of Mn. [1]

c) Most of the transition elements show paramagnetic behavior. why? [1]

d)  $\text{Fe}^{3+}$  is more stable than  $\text{Fe}^{2+}$ . why? [1]

e) Explain why  $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$  ion is coloured? [1]

Set B

1. a.  $5 \times 10^{-5}$  kg

2. b. NaOH

3. a. stable

4. d.  $\text{Cl}^-$

5. c. +4

6. a.  $\text{H}_2\text{O}$

7. b. low boiling point of zinc

8. a.  $3^\circ > 2^\circ > 1^\circ$

9. b. nucleophilic substitution

10. b. 3

11. a. carbocation

6. a) Starting from iodoethane, how would you obtain. [2]  
i) Methoxyethane ii) Bromoethane

- b) A sweet-smelling organic compound 'A' is slowly oxidized by air in the presence of light to give a highly poisonous gas. On heating 'A' with silver powder, it forms a gaseous product 'B'. 'B' can be also produced by the action of water on calcium carbide. Identify 'A' and 'B' and write the chemical equations of the reaction involved. [3]

7. a) Write the reaction for preparation of trichloromethane from ethanol. [3]

- b) Give an example of: [1+1]  
i) Reimer - Tiemann's reaction  
ii) Williamson ether synthesis.

8. An organic compound (A) which is the functional isomer of butan-1-ol and it is commonly called 'sulphuric ether'.

- a) Identify compound 'A' and write its IUPAC name. [1]  
b) What happens when the compound 'A' is treated with cold and conc. HCl? [1]  
c) What happens when nitroalkane is reduced [1+1+1]  
i) in acidic medium?  
ii) in neutral medium?  
iii) by catalytic reduction?

**GROUP 'C' [3+8+24]**

**Long Questions**

9. a) Define common ion effect with a suitable example. [2]  
b) Discuss about the application of common ion effect and solubility product principle. [3]  
c) 85 ml of 0.1 M  $H_2SO_4$  is treated with 0.1N KOH solution until the acid is one-fourth neutralized. Calculate the concentration of mixture solution in terms of salt formed and in terms of acid left. [3]
10. a) Write the action of heat on white vitriol. [2]

- b) What happens when zinc is exposed to moist air? [1]  
c) What is lithopone? How is it formed? [1]  
d) Write down the process of extraction of zinc with necessary diagrams. [4]
11. a) How is chlorobenzene prepared from  
i) Benzene ii) Benzene diazonium chloride? [1+1]  
b) Chlorobenzene is less reactive than chloroethane towards nucleophilic substitution reaction, why? [2]  
c) Write the possible structural isomers of  $C_6H_5Br$  with their IUPAC names and common names. Which isomer will have lowest boiling point and why? [3+1]

"Best of Luck"

KMC Note

6. a) Starting from iodoethane, how would you obtain.

- i) Methoxyethane [2]  
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[3]

8. a) Write the reaction for preparation of trichloromethane from ethanol.

[3]

b) Give an example of:

[1+1]

i) Reimer - Tiemann's reaction.

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9. An organic compound (A) which is the functional isomer of butan-1-ol and it is commonly called 'sulphuric ether'.

a) Identify compound 'A' and write its IUPAC name. [1]

b) What happens when the compound 'A' is treated with cold and conc. HCl? [1]

c) What happens when nitroalkane is reduced [1+1+1]

i) in acidic medium?

ii) in neutral medium?

iii) by catalytic reduction?

**GROUP 'C' [3+3+24]**

#### Long Questions

9. a) Define common ion effect with a suitable example. [2]

b) Discuss about the application of common ion effect and solubility product principle. [3]

10. 5 ml of 0.1 M  $H_2SO_4$  is treated with 0.1N KOH solution until the acid is one-fourth neutralized. Calculate the concentration of mixture solution in terms of salt formed and in terms of acid left. [3]

10. a) Write the action of heat on white vitriol. [2]

b) What happens when zinc is exposed to moist air? [1]

c) What is lithopone? How is it formed? [1]

d) Write down the process of extraction of zinc with necessary diagrams. [4]

11. a) How is chlorobenzene prepared from

i) Benzene [1+1]  
ii) Benzene diazonium chloride? [1+1]

b) Chlorobenzene is less reactive than chloroethane towards nucleophilic substitution reaction, why? [2]

c) Write the possible structural isomers of  $C_4H_9Br$  with their IUPAC names and common names. Which isomer will have lowest boiling point and why? [3+1]

"Best of Luck"

KMC Note